A Decision Support System for Business Continuity and Critical Service Recovery Prioritization

A Master's Project Submitted in Partial Fulfillment of the Requirements for the Master's Degree in Information Systems

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ABSTRACT

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الخلاصة

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LIST OF ABBREVIATIONS

If appropriate, list any abbreviations in alphabetic order used in the thesis.

CDF	Cumulative Distribution Function
KSU	King Saud University

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ACKNOWLEDGEMENTS

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Chapter 1: Introduction

1.1 Motivation and Scope

Organizations are required to keep delivering their essential services even when operations are disrupted, whether the disruption reaches their activities, infrastructure, information, or the external parties they depend on. How serious an interruption turns out to be depends on more than the incident itself. It also depends on the importance of the affected service and on how long that service stays unavailable. An interruption that can be absorbed at first may become unacceptable once its effects on service users, operations, financial obligations, regulatory requirements, and institutional reputation begin to accumulate. For this reason, business continuity management calls on organizations to identify their priority services, to understand how the impacts of an interruption grow over time, and to set recovery requirements they can defend before any disruption takes place [1], [2].

Business continuity management (BCM) offers a structured way to prepare for disruptions and to maintain or restore prioritized activities and services. Within it, business impact analysis (BIA) plays a central part: it studies the consequences of an interruption, establishes how important services and activities are relative to one another, and feeds into recovery planning. Two recovery parameters matter most for this research. The maximum tolerable period of disruption (MTPD) is the duration beyond which the consequences of an interruption become unacceptable, and the recovery time objective (RTO) is the target time for resuming a service or activity. The two are related but serve different purposes, and each has to be set with reference to the impacts the organization has assessed and the recovery requirements it has identified [1], [2].

Assessing the consequences of an interruption is inherently a matter of time. Services differ in how quickly the impacts of an interruption become serious, and organizations may also differ in the criteria they use to judge those impacts. The business continuity guidance of the Digital Government Authority (DGA), for instance, presents impact-based classifications and recovery targets for government digital

platforms, and at the same time recognizes that the time intervals used for assessment should follow the sensitivity of the activities concerned [1]. These principles point toward an approach in which each organization defines its own assessment periods, impact criteria, acceptable thresholds, and matching recovery rules, rather than assuming that one fixed configuration suits everyone.

This project proposes Time-Dependent Recovery Prioritization (TDRP), a configurable rule-based methodology, and Masar, a web-based decision support system (DSS) that implements it. The approach starts from business services and their BIA results, applies explicit rules to their time-dependent impact assessments, and derives criticality, recovery requirements, and relative recovery priorities. The dependencies that support each service are then related to its criticality, and the system presents the outcome on an operational dashboard. Its purpose is to help responsible personnel interpret and use continuity information, not to replace their professional judgment or to take on responsibility for operational recovery decisions.

The scope of the research covers the design, implementation, and evaluation of the TDRP methodology within Masar. The central concern is the service-level assessment and prioritization process. Supporting dependencies, that is, the elements and resources on which service delivery relies, are considered only to the extent that their relationships help explain how services are delivered and what they require for recovery.

1.2 Problem Statement and Significance for Information Systems

Deciding on recovery priorities takes more than recognizing that a service is important. An organization has to be able to explain how the assessed impacts of an interruption justify a service's criticality, how those impacts relate to acceptable disruption durations and recovery targets, and how the resulting priorities should be read alongside the dependencies that the service needs in order to be delivered. National business continuity guidance names BIA, recovery requirements, and dependencies as important components of continuity management [1], [2]. Yet these principles do not lay down a single mandatory computational procedure that every organization must follow to turn successive impact assessments into an MTPD, an RTO,

and service recovery priorities. Nor can the thresholds or assessment intervals of one organization be taken as suitable for any other.

The research problem is therefore the need for a clear, configurable, and reproducible method that translates time-dependent impact assessments of business services into recovery requirements and relative recovery priorities, while keeping an explicit connection between the assessed services and the dependencies that support their delivery. Such a method should make its decision rules visible and let different organizations configure them according to their approved criteria. Its results should also be traceable to the assessments underlying them. In addition, it should distinguish the priority assigned to a service from the importance given to a supporting dependency because other services rely on it. The project does not assume that continuity practices are absent today or that organizations lack information. It addresses, instead, the formalization and implementation of one particular service-first prioritization procedure.

The problem matters to information systems because it concerns the transformation of organizational assessments into consistent and explainable decision-support outputs. A DSS can represent services, time-based impacts, recovery parameters, configurable rules, and dependency relationships in a single coherent information model, and it can apply the same declared logic to comparable inputs. This makes it possible to inspect how a given priority was obtained, to examine what happens when an approved rule configuration is changed, and to present the resulting information to continuity practitioners. Designing and evaluating a purposeful artifact of this kind is in line with design-science research, which is concerned with building and assessing artifacts that address identified organizational problems [3], [4].

1.3 Importance of the Problem and the Big Picture

Business continuity is an organizational capability and not merely a technical recovery activity. A disruption can affect the availability of public or private services, the obligations an organization must meet, the experience of its beneficiaries, and its capacity to keep essential operations running. The Saudi National Framework for Risk, Emergency and Business Continuity Management treats business continuity as one part

of a broader framework, relates it to the ability of an organization to withstand disruption, and calls for its alignment with the organization's own strategy [1]. The DGA likewise stresses sustained service delivery and the alignment of recovery arrangements with organizational objectives [2]. Seen from this angle, prioritization is important because organizations need to determine which service outcomes are the most time-sensitive and to let those priorities inform their recovery efforts.

A service-oriented view also offers a principled way of treating supporting dependencies. Services are delivered through interacting activities and enabling resources, and the interruption of a shared dependency can affect more than one service. The criticality of a supporting dependency should therefore be read in relation to the services it enables, and not inferred from its own characteristics alone. In the proposed methodology, service criticality is established first. The relevant dependency relationships are recorded next, and a stated rule links each supporting dependency to the criticality of the service level. Where several services share a dependency, the methodology has to define how their classifications are combined. Taking the highest criticality among the supported services is a rule proposed here for examination and validation. It is not a general requirement that the reference frameworks can be said to impose.

Another concern is how to balance standardization against organizational flexibility. Consistency calls for a common method of applying rules, whereas practical use calls for organizations to choose impact scales, assessment time frames, and recovery thresholds that fit their own contexts. A configurable rule-based methodology can meet both needs by standardizing the form and execution of the decision logic without requiring identical parameter values from every organization. The contribution anticipated in this sense is not a replacement for existing BCM standards or for the expertise of practitioners. It is a transparent procedure for interpreting BIA results and expressing recovery priorities. Research on business impact analysis and on integrated continuity planning gives a broader basis for studying the link between organizational impacts and recovery planning [5], [6].

1.4 Objectives and Proposed Solution

The aim of this project is to design, implement, and evaluate a configurable rule-based methodology that derives recovery requirements and prioritizes critical business services from time-dependent impact assessments. The methodology, TDRP, is the main research contribution. Masar is the proposed web-based DSS in which the methodology will be configured, applied, and evaluated, and through which its results will be shown to authorized users. More specifically, the project pursues five objectives. The first is to derive service criticality, MTPD, and RTO from time-dependent impact assessments through configurable rules, and the second is to establish relative recovery priorities from criticality and recovery time requirements. The third is to associate services with their supporting dependencies and to assign criticality to those dependencies by a documented rule. The fourth is to implement the methodology in Masar, with an operational dashboard that presents its results, and the fifth is to evaluate the correctness, consistency, and traceability of the methodology and its implementation.

The methodology will represent BIA findings at the service level across successive assessment periods and the relevant impact dimensions. Following the criteria and decision rules defined by the organization, it will determine service criticality, identify the applicable MTPD, and establish the corresponding RTO. It will also document how temporal assessments, impact thresholds, and recovery-time mappings are interpreted, including how incomplete data are handled and what happens when an assessed impact does not reach a threshold. Organizations will be able to configure the permitted parameters so that they reflect their approved continuity requirements.

From these service-level results, TDRP will set relative recovery priorities through explicit rules that take criticality and recovery-time requirements into account. The approach will then associate each assessed service with the dependencies needed to deliver it. A supporting dependency will inherit criticality from the service or services it supports according to a documented rule. For a dependency shared by several services, the proposed default is the highest criticality among them, subject to evaluation. The purpose of this association is to preserve the business reasoning behind the importance of supporting dependencies. A ranking of services by priority is not to be equated with an automatically executable sequence of recovery operations.

Masar will put the methodology into practice through an integrated data model and rule processing functions. Authorized users will be able to maintain services, impact assessments, configuration settings, recovery requirements, and the relevant dependency relationships. An operational dashboard will present the resulting service classifications, MTPD and RTO values, recovery priorities, and linked supporting dependencies. During an actual interruption, users can consult the recorded information on the affected services to see their established priority and dependencies. The dashboard will show the outputs of the methodology together with their context. It will not simulate incidents, infer operational actions, recommend recovery procedures on its own, or make or carry out recovery decisions.

The research will evaluate the methodological rules and their implementation. The evaluation will look at the consistency and reproducibility of the outputs, the behavior of the method under different valid configurations, the correctness of the derived MTPD and RTO values against specified test cases, the treatment of shared dependencies, and the traceability of displayed priorities to the impact assessments and rules they come from. The intended deliverables are a documented TDRP methodology, an implemented Masar prototype, and an evaluation of their correctness and usefulness as a basis for structured decision support. How effective the approach is in real incidents lies beyond what a prototype-level evaluation alone can establish.

Chapter 2: Literature Review

The literature review chapter should demonstrate a thorough knowledge of the area and provide arguments to support the study focus.

Chapter 3: Methodology/ Research Design

This chapter of the proposal should outline the design and methodology of your research. The basis for the choice of research method should be whether it will help you answer your research question(s).

Chapter 4: Results and Discussions

Note: Results and Discussions can be presented in a single chapter or in two different chapters depending on the length of your thesis.

Chapter 5: Analysis/ Discussions

Note: Similarly, Results and Discussions can be presented in a single chapter or in two different chapters depending on the length of your thesis.

Chapter 6: Conclusions

This chapter contains conclusions, limitations, and recommendations – so what is the theory?

References

[1] General Secretariat of the National Risk Council, “National Framework for Risk, Emergency and Business Continuity Management: Business Continuity Management Section.” [Online]. Available: https://riskbcm.com/gsnrc_standards/

[2] Digital Government Authority, “Guideline for Risk Management and Business Continuity for Digital Government: Business Continuity Management System,” Digital Government Authority, Saudi Arabia, DGA-1-2-5-231, Jan. 2025. [Online]. Available: https://dga.gov.sa/ar/regulatory-documents/guideline-for-risk-management-and-business-continuity-for-digital-government

[3] A. R. Hevner, S. T. March, J. Park, and S. Ram, “Design Science in Information Systems Research,” *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004, doi: 10.2307/25148625.

[4] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A Design Science Research Methodology for Information Systems Research,” *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007, doi: 10.2753/MIS0742-1222240302.

[5] S. A. Torabi, H. Rezaei Soufi, and N. Sahebjamnia, “A New Framework for Business Impact Analysis in Business Continuity Management (with a Case Study),” *Safety Science*, vol. 68, pp. 309–323, 2014, doi: 10.1016/j.ssci.2014.04.017.

[6] N. Sahebjamnia, S. A. Torabi, and S. A. Mansouri, “Integrated Business Continuity and Disaster Recovery Planning: Towards Organizational Resilience,” *European Journal of Operational Research*, vol. 242, no. 1, pp. 261–273, 2015, doi: 10.1016/j.ejor.2014.09.055.

Appendices

Appendix A

Title

Start each appendix on a new page. Place appendices in the same order as they are referred to in the body of the thesis. That is, the first appendix referred to should be Appendix A, the second appendix referred to should be Appendix B, and so on. Appendix formatting can be different to the main document.

Appendix B

List of Publications

Add the first page of each research papers, journal articles, and conference proceedings that have arisen from the research being presented in your thesis.